

Improving Downstream Task Performance in Multilingual LLMs by Intervening Language and Task Specific Neurons

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by

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Approval

This BPS entitled "**Improving Downstream Task Performance in Multilingual LLMs by Intervening Language and Task Specific Neurons**" by **Soumen Kumar Mondal**, Roll No. 23m2157, is approved for the degree of *Master of Science by Research in Data Science and Artificial Intelligence*.

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Declaration

I declare that this written submission represents my ideas in my own words. Where others' ideas and words have been included, I have adequately cited and referenced the original source. I declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated, or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will cause disciplinary action by the Institute and can also evoke penal action from the source which has thus not been properly cited or from whom proper permission has not been taken when needed.

Soumen Kumar Mondal

Date: December 27, 2024

Place: Mumbai, India

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Date: December 27, 2024

Place: Mumbai, India

Abstract

This thesis investigates methodologies for identifying and utilizing language- and task-specific neurons in multilingual large language models (LLMs) to improve performance on multilingual downstream tasks, particularly for low-resource languages. The research examines how language-specific information is represented in various LLM architectures, including Llama3.1, Bloomz, and Aya models. Techniques such as activation probability entropy and LoRA-based fine-tuning methods were applied to identify language- and task-specific neurons using datasets like Wikipedia and XNLI. Intervention strategies, such as replacing neuron activations with language-specific statistical activations, were explored to determine their effect on task performance. Experiments involving neuron-specific fine-tuning and neuron freezing were also conducted to study the relationship between architectural modifications and language representation.

The study further analyzes perplexity distributions and neuron activation maps across different languages and models, providing insights into how multilingual datasets influence the internal mechanisms of LLMs. Experimental setups were developed to assess the effectiveness of fine-tuning and intervention methods for improving natural language inference (NLI) performance in low-resource languages without requiring direct training data. By addressing existing gaps in language-specific neuron identification, this work contributes to a better understanding of how LLMs can be tailored to achieve performance improvement across diverse languages. The results provide a deeper understanding of the interaction between LLM architecture and multilingual capabilities, particularly for underrepresented languages.

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Abbreviations

AFT	A ccelerated F ailure T ime
ANOVA	A nalysis of V ariance
BLR	B inary L ogistic R egression

Symbols

μ	Location parameter
λ	Scale parameter
β_0	Null model parameter
$\hat{\beta}$	Fitted parameter

Chapter 1

Introduction

1.1 Introduction

The emergence of large language models (LLMs) has transformed natural language processing (NLP), enabling remarkable advancements across a wide range of linguistic tasks. Despite their success, challenges persist in ensuring comparable performance for low-resource languages, which often lack sufficient representation in pretraining datasets. This thesis addresses these challenges by investigating the representation of language-specific information within LLMs and exploring methods to identify and utilize language-specific neurons. By focusing on multilingual tasks such as natural language inference (NLI), this research seeks to better understand how language-specific information is encoded and how it can be effectively utilized to improve performance, particularly for underrepresented languages.

The research builds on recent advancements in neuron analysis and fine-tuning techniques, using models such as Llama3.1, Bloomz, and Aya to explore the relationship between architectural choices, pretraining datasets, and language-specific representations. By employing methods such as activation probability entropy and LoRA-based fine-tuning, the study systematically identifies language-specific neurons and evaluates their role in downstream tasks. Through detailed experiments on layer-specific fine-tuning, freezing, and intervention strategies, the thesis aims to provide a comprehensive understanding of the mechanisms underlying multilingual capabilities in LLMs.

1.2 Motivation

In the era of multilingual LLMs, achieving consistent performance across diverse languages remains a significant challenge. High-resource languages benefit from extensive pretraining data and resources, leading to superior performance on downstream tasks, while low-resource

languages often lag behind. Addressing this imbalance is essential for ensuring that the advancements in NLP benefit speakers of all languages, regardless of the resources available for their linguistic representation.

Understanding the internal mechanisms of LLMs, specifically how they represent language-specific information, offers a way to addressing these challenges. By identifying language-specific neurons and their roles in encoding linguistic knowledge, we can develop targeted interventions to improve performance for low-resource languages. This research is driven by the need to close the performance gap, advance the understanding of multilingual model architectures, and contribute to the development of fair and inclusive NLP systems.

1.3 Problem Statement

While LLMs exhibit strong multilingual capabilities, their performance across languages is often inconsistent, with low-resource languages showing suboptimal results compared to high-resource counterparts. This disparity can be attributed to the lack of sufficient representation for low-resource languages during pretraining and the challenge of effectively isolating and utilizing language-specific information within these models. Existing methods for identifying language-specific neurons often rely on arbitrary thresholds or lack generalizability across different architectures and datasets. Moreover, the impact of architectural modifications, such as layer-specific fine-tuning or neuron freezing, on language-specific representation and downstream performance remains underexplored.

This research aims to address these limitations by developing systematic approaches to identify and analyze language-specific neurons in LLMs. By focusing on techniques such as neuron activation probability and LoRA-based fine-tuning, the study seeks to bridge the gap in understanding how language-specific information is encoded. The ultimate goal is to improve task performance for low-resource languages without requiring extensive annotated data, thereby contributing to the application of LLMs in multilingual settings.

Chapter 2

Literature Review

2.1 Language-Specific Neurons in Multilingual Models

A significant body of work has focused on understanding and isolating language-specific neurons within large language models (LLMs). The paper *Language-Specific Neurons in Multilingual Language Models* ([ArXiv:2404.02431](#)) investigates techniques for identifying neurons that are specific to certain languages and evaluates their impact on tasks such as machine translation and multilingual understanding. Building on this, our research extends the idea by evaluating alternative methods such as replacing median activations with layer-specific median or 75th percentile activations and exploring whether language-specific neuron sets vary across datasets, tasks, and model layers.

The reproduction of results from the *Language-Specific Regions* paper ([ArXiv:2402.16438](#)) further validated the connection between perplexity changes and language specificity in LLMs. The study provided insights into how replacing or interpolating neuron activations affects task performance, particularly for low-resource languages. These findings underline the importance of understanding neuron activity patterns for improving multilingual tasks. For instance, experiments demonstrated that full replacement of language-specific neurons may degrade performance, while linearly interpolating activation values can balance task performance and neuron specificity.

2.2 Mechanistic Interpretability and Multilinguality

The broader context of neuron analysis in multilingual LLMs intersects with mechanistic interpretability, as explored in works such as the *SoLU Activation Function* ([SoLU paper](#)) and *Tracing the Roots of Facts in Multilingual Models* ([ArXiv:2403.05189](#)). The former highlights methods to reduce polysemanticity in neurons, which could aid in isolating neurons responsible for specific linguistic features. Meanwhile, the latter paper delves into probing techniques for

factual association in multilingual models, suggesting that neuron activity patterns can vary significantly depending on the dataset and task.

This thesis builds on these foundational ideas by experimenting with probing methods such as L1 and L2 norms and designing novel interventions in neuron activations. The literature also emphasizes how architectural changes, such as adding adapter layers or modifying attention heads, can enhance model performance on multilingual tasks. The exploration of neuron activity patterns for cross-lingual transfer, fact probing, and multilingual perplexity maps adds depth to the understanding of language representation in LLMs.

2.3 Applications to Low-Resource Languages

Addressing performance disparities for low-resource languages is a recurring theme across the literature. Papers such as *Locating and Editing Factual Associations in GPT* and *Discovering Salient Neurons in Deep NLP Models* highlight methodologies for identifying and modifying neurons to improve downstream task performance. This aligns with our research objectives, particularly in the context of improving natural language inference (NLI) and translation tasks for low-resource languages. Techniques such as activation replacement, layer freezing, and fine-tuning specific neurons have shown promise in achieving consistent task performance across diverse linguistic settings.

Chapter 3

Methodology

3.1 Definition of Neuron

In the context of multi layer perceptrons (MLPs) in large language models (LLMs), a *neuron* refers to a unit of computation within the MLP module. Specifically, consider the i^{th} layer of a Transformer architecture, where the input is a hidden state vector $\mathbf{H}^{i-1} \in \mathbb{R}^{T \times d}$, derived from the previous layer.

Hidden State Representation: The output of the multi-head self-attention (MHA) mechanism serves as input to the MLP and is given as:

$$\hat{\mathbf{H}}^i = \text{Attn}(\mathbf{H}^{i-1}\mathbf{W}_q^i, \mathbf{H}^{i-1}\mathbf{W}_k^i, \mathbf{H}^{i-1}\mathbf{W}_v^i) \cdot \mathbf{W}_o^i, \quad (3.1)$$

where $\mathbf{W}_q^i, \mathbf{W}_k^i, \mathbf{W}_v^i, \mathbf{W}_o^i \in \mathbb{R}^{d \times d}$ are the trainable parameters of the self-attention module, and $\text{Attn}(\cdot)$ denotes the scaled dot-product attention operation.

MLP Transformation: The MLP module takes $\hat{\mathbf{H}}^i$ as input and transforms it as follows:

$$\mathbf{h}^i = \text{act_fn}(\hat{\mathbf{H}}^i \mathbf{W}_1^i) \cdot \mathbf{W}_2^i, \quad (3.2)$$

where $\mathbf{W}_1^i \in \mathbb{R}^{d \times 4d}$ and $\mathbf{W}_2^i \in \mathbb{R}^{4d \times d}$ are learnable projection matrices, and $\text{act_fn}(\cdot)$ represents a non-linear activation function such as GELU or SiLU.

Neuron: Within the MLP, the j^{th} neuron of the i^{th} layer corresponds to the j^{th} row of the intermediate activation matrix:

$$\mathbf{h}_j^i = \text{act_fn}((\hat{\mathbf{H}}^i \mathbf{W}_1^i)_j), \quad (3.3)$$

where $(\hat{\mathbf{H}}^i \mathbf{W}_1^i)_j$ represents the pre-activation value of the j^{th} neuron.

Generalized Neuron Formulation: Recent architectures such as LLaMA utilize Gated Linear Units (GLU) to enhance MLP performance, modifying the MLP computation as:

$$\mathbf{h}^i = [\text{act_fn}(\hat{\mathbf{H}}^i \mathbf{W}_1^i) \otimes (\hat{\mathbf{H}}^i \mathbf{W}_3^i)] \cdot \mathbf{W}_2^i, \quad (3.4)$$

where $\mathbf{W}_3^i \in \mathbb{R}^{d \times 4d}$ is an additional projection matrix, and $\otimes$ denotes element-wise multiplication. This formulation introduces gating to the MLP, allowing for more expressive transformations. Figure 3.1 shows the architecture of the MLP module.

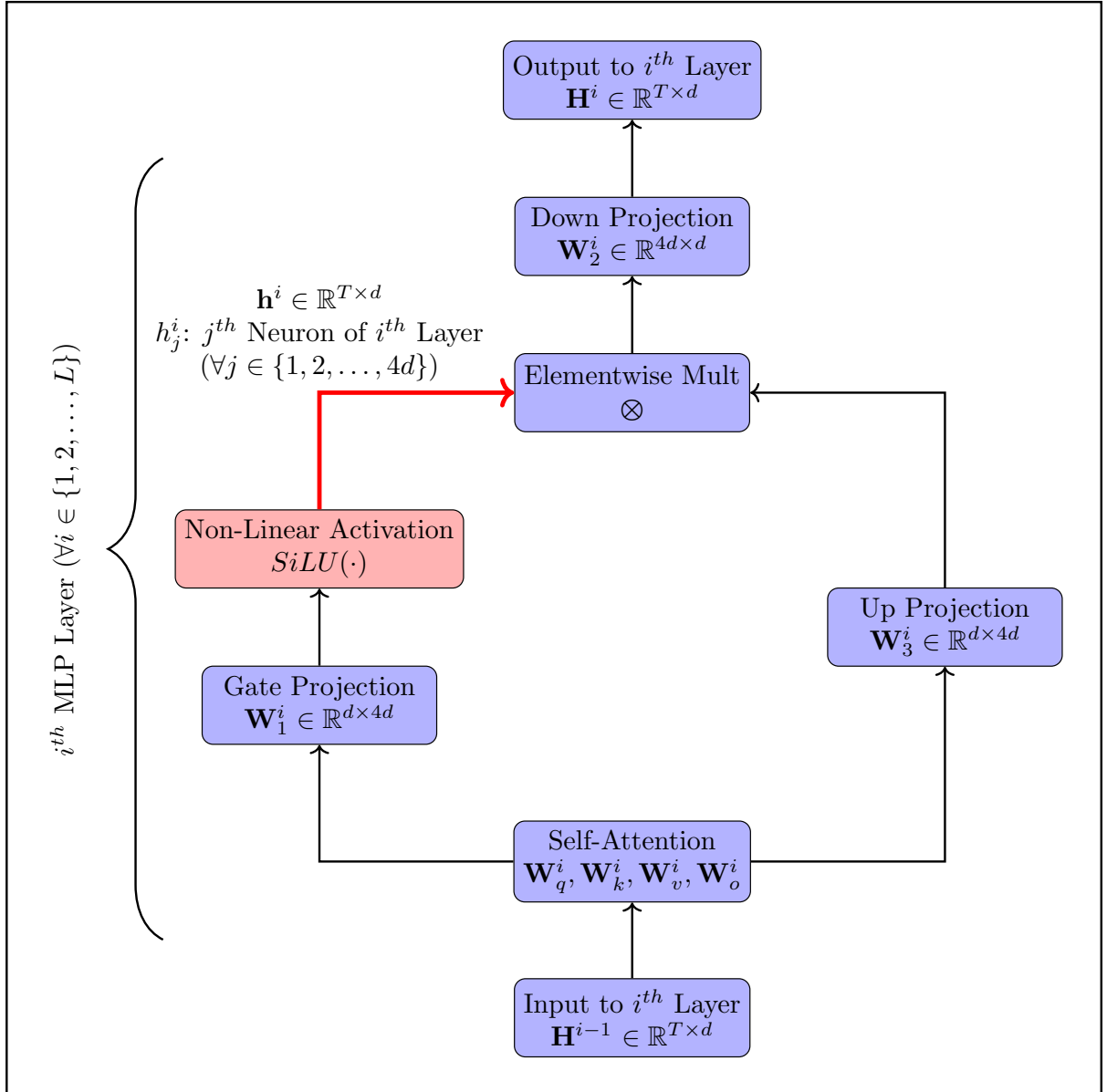


FIGURE 3.1: Architecture of MLP module in LLaMA model

3.2 Dataset for Neuron Identification

For the purpose of identifying language-specific neurons, we utilize a subset of the Wikipedia dataset spanning 15 languages:

$$\{\text{en, fr, es, vi, id, ja, zh, bn, hi, ta, te, mr, ur, kn, ml, pa}\}.$$

Each language's dataset is preprocessed and passed through pretrained large language models (LLMs). The activations are then collected from the MLP modules of the models.

Dataset Description: The Wikipedia dataset for each language contains a sequence of tokenized text data. Each tokenized sequence has a maximum length of $T_{\max} = 512$. For every sequence, we compute the activations from the MLP module, followed by averaging the activations at each neuron position across all sequences. This provides a language-specific profile for every neuron (i, j) in the model.

Activation Collection: Given a pretrained LLM, let the MLP activation for the i^{th} layer and the j^{th} neuron in a specific language be represented by $h_{i,j}^t \in \mathbb{R}$, where t indexes the token positions in a sequence of length T . The average activation of a neuron across all token positions for a given sequence is computed as:

$$\bar{h}_{i,j} = \frac{1}{T} \sum_{t=1}^T h_{i,j}^t, \quad (3.5)$$

where $h_{i,j}^t$ is the neuron activation corresponding to the t^{th} token.

Language-Specific Activations: For each language l in the dataset, we compute the average activation of each neuron (i, j) over all sequences S_l in the dataset:

$$H_{i,j}^l = \frac{1}{|S_l|} \sum_{s \in S_l} \bar{h}_{i,j}^{(s)}, \quad (3.6)$$

where $H_{i,j}^l$ represents the average activation of the j^{th} neuron in the i^{th} layer for language l , and $|S_l|$ denotes the number of sequences for language l .

3.3 Statistics of Neuron Activation

The statistics of neuron activation are precomputed to analyze language-specific patterns in the MLP layers of the pretrained large language models (LLMs). These statistics include mean, standard deviation, and quantiles (e.g., 5th, 10th, 25th, 50th, 75th, 90th, and 95th percentiles) of the neuron activations for each language in the dataset. The computations are performed across all layers and neurons within the MLP modules.

Activation Mean and Standard Deviation: The mean activation of a neuron (i, j) in the t^{th} token position is computed as:

$$\mu_{i,j} = \frac{1}{T} \sum_{t=1}^T h_{i,j}^t, \quad (3.7)$$

where $h_{i,j}^t$ is the activation of the (i, j) neuron for the t^{th} token, and T is the total number of tokens in a sequence. Similarly, the standard deviation of the activations is calculated as:

$$\sigma_{i,j} = \sqrt{\frac{1}{T} \sum_{t=1}^T (h_{i,j}^t - \mu_{i,j})^2}. \quad (3.8)$$

Quantiles of Activation: Quantile-based statistics provide insights into the distribution of activations for each neuron. The q -quantile activation value, denoted as P_q , is defined as:

$$P_q(h_{i,j}) = \inf \{x \in \mathbb{R} : F_{h_{i,j}}(x) \geq q\}, \quad (3.9)$$

where $F_{h_{i,j}}(x)$ is the cumulative distribution function (CDF) of the activations for neuron (i, j) . Commonly computed quantiles include:

$$P_{50}, P_{75}, P_{90}, P_{95}, P_{25}, P_{10}, P_5.$$

Language-Specific Aggregation: For a given language l and all sequences in the dataset, the mean and other statistics are aggregated across all token positions and batches. Let S_l denote the set of sequences for language l , and B represent the batch size. The aggregated mean activation for neuron (i, j) is computed as:

$$\bar{\mu}_{i,j}^l = \frac{1}{|S_l|} \sum_{b=1}^B \sum_{s \in S_l} \mu_{i,j}^{(s,b)}, \quad (3.10)$$

where $\mu_{i,j}^{(s,b)}$ is the mean activation of neuron (i, j) for the s^{th} sequence in the b^{th} batch.

3.4 Methods for Neuron Identification

3.4.1 LAPE

Language Activation Probability Entropy (LAPE) is a metric designed to identify neurons that exhibit language-specific behavior across multiple languages. This metric quantifies the entropy of normalized activation probabilities for each neuron across the given set of languages.

Step 1: Compute Activation Probabilities. For a pretrained large language model with L layers and $4d$ neurons per layer, the activation of a neuron (i, j) in response to a sequence $X^l \in \mathbb{R}^{N \times T}$ (where N is the number of sequences, and T is the number of tokens per sequence)

for a language l is denoted as $h_{i,j}^l$. The unnormalized activation probability for neuron (i, j) is given by:

$$p_{i,j}^l = \mathbb{E} [\mathbb{I}(h_{i,j}^l > 0)], \quad (3.11)$$

where $\mathbb{I}$ is the indicator function that returns 1 if the activation $h_{i,j}^l$ is greater than zero, and 0 otherwise. The expectation is computed over all tokens in all sequences for language l .

Step 2: Normalize Activation Probabilities. The normalized activation probability is computed across k languages to ensure a valid probability distribution:

$$P_{i,j}^l = \frac{p_{i,j}^l}{\sum_{l=1}^k p_{i,j}^l}, \quad \forall i \in \{1, \dots, L\}, \forall j \in \{1, \dots, 4d\}. \quad (3.12)$$

Step 3: Compute Language Activation Probability Entropy. The entropy of the normalized activation probabilities for each neuron (i, j) is computed as:

$$\text{LAPE}(i, j) = - \sum_{l=1}^k P_{i,j}^l \log P_{i,j}^l, \quad \forall i \in \{1, \dots, L\}, \forall j \in \{1, \dots, 4d\}. \quad (3.13)$$

This measures the uncertainty in the activation distribution of a neuron across the k languages.

Step 4: Apply Threshold and Select Neurons. To identify neurons with significant language-specific behavior, a threshold is defined based on the 95th percentile of the unnormalized activation probabilities:

$$\text{Threshold} = \text{Quantile}_{0.95}(p_{i,j}^l | \forall i, j, l). \quad (3.14)$$

A neuron (i, j) is considered for language-specific analysis only if:

$$\exists l \in \{1, \dots, k\} \text{ such that } p_{i,j}^l > \text{Threshold}. \quad (3.15)$$

Step 5: Select Top Neurons Based on LAPE. The top m neurons with the lowest LAPE values are selected for language-specific analysis, where m is defined as:

$$m = \frac{0.01 \times 4d \times L}{100}. \quad (3.16)$$

This ensures that only the most language-specific neurons are retained for further interpretation.

Step 6: Identify Language-Specific Neuron Sets. For each neuron (i, j) in the selected set, the language-specific set is defined as:

$$\text{Lang_Spec_Set}(i, j) = \{l | P_{i,j}^l > \text{Threshold}\}. \quad (3.17)$$

A neuron is considered specific to a subset of languages if:

$$\text{Lang_Spec_Set}(i, j) \subseteq \{1, 2, \dots, k\}. \quad (3.18)$$

3.4.2 Activation Based Methods

Absolute Activation: The absolute activation metric measures the mean absolute activation of each neuron (i, j) in the i^{th} layer across all tokens in a sequence for a language l . This is computed as:

$$r_{i,j}^l = \mathbb{E} [|h_{i,j}^l|], \quad (3.19)$$

where $h_{i,j}^l$ is the activation of neuron (i, j) for language l , and the expectation is taken over all tokens and sequences. This metric identifies neurons with consistently high absolute activations, which are indicative of their overall importance in representing language features.

Activation Probability Zero: The activation probability zero metric measures the fraction of times a neuron (i, j) has a positive activation for a language l . It is defined as:

$$r_{i,j}^l = \mathbb{E} [\mathbb{I}(h_{i,j}^l > 0)], \quad (3.20)$$

where $\mathbb{I}$ is the indicator function that returns 1 if $h_{i,j}^l > 0$, and 0 otherwise. This metric captures the frequency of a neuron's activation, providing insights into its contribution to language-specific representations.

Activation Probability Mean: The activation probability mean measures the probability of a neuron's activation exceeding the mean activation value for its layer, $\mu_{i,j}^l$, for a language l . It is computed as:

$$r_{i,j}^l = \mathbb{E} [\mathbb{I}(h_{i,j}^l > \mu_{i,j}^l)], \quad (3.21)$$

where $\mu_{i,j}^l$ represents the average activation of neuron (i, j) in the i^{th} layer for language l . This metric identifies neurons with activations that are consistently above the average, highlighting neurons that are highly responsive to language-specific features.

Activation Probability 90p: The activation probability 90p measures the probability of a neuron's activation exceeding the 90th percentile activation value for its layer, $\text{P90}_{i,j}^l$, for a language l . This is given by:

$$r_{i,j}^l = \mathbb{E} [\mathbb{I}(h_{i,j}^l > \text{P90}_{i,j}^l)], \quad (3.22)$$

where $\text{P90}_{i,j}^l$ represents the 90th percentile of the activations of neuron (i, j) in the i^{th} layer for language l . This metric identifies neurons that exhibit rare but significant activations, which may correspond to specialized processing for certain linguistic features.

Language-specific neurons are identified based on activation values computed across different languages. For each language, activations are collected, and neurons that exhibit the highest relevance are selected using the following process.

Step 1: Compute Activation Relevance. For each language $l \in \{1, 2, \dots, k\}$, compute the activation relevance values for all neurons (i, j) in the pretrained model. Let the relevance for neuron (i, j) in the i^{th} layer for language l be denoted as $r_{i,j}^l$. This relevance is calculated using one of the activation probability methods, such as:

$$r_{i,j}^l = \mathbb{E} [\mathbb{I}(h_{i,j}^l > 0)], \quad (3.23)$$

or other methods such as absolute activation or percentile-based thresholds, depending on the desired analysis.

Step 2: Aggregate Relevance Across Layers and Neurons. For each language l , the computed relevance values $r_{i,j}^l$ are aggregated across all layers and neurons. The total number of layers is denoted by L , and the dimensionality of the neurons in each layer is $4d$. The relevance tensor for language l is represented as:

$$R^l = \{r_{i,j}^l \mid i \in \{1, \dots, L\}, j \in \{1, \dots, 4d\}\}. \quad (3.24)$$

Step 3: Select Top Neurons. For each language l , the top m neurons with the highest relevance values are selected. The value of m is determined as a small fraction of the total number of neurons in the model:

$$m = \frac{0.125 \times L \times 4d}{100}. \quad (3.25)$$

The top m neurons are identified by sorting the relevance values in R^l and selecting the indices of the m largest values:

$$\text{Top}_m(R^l) = \text{argsort_descending}_m \{r_{i,j}^l \mid \forall i, j\}. \quad (3.26)$$

Step 4: Map Neurons to Layers and Positions. The indices of the top m neurons are mapped back to their corresponding layers and positions. For each language l , the selected neurons are represented as:

$$N^l = \{(i, j) \mid (i, j) \in \text{Top}_m(R^l)\}, \quad (3.27)$$

where N^l is the set of top m neurons for language l .

Step 5: Store Language-Specific Neuron Sets. The selected neurons for each language are stored in a dictionary, mapping each language l to its corresponding set of neurons:

$$\text{Lang_Neuron_Set} = \{l \mapsto N^l \mid l \in \{1, 2, \dots, k\}\}. \quad (3.28)$$

This process ensures that neurons with the highest language-specific relevance are identified for each language. These neurons can then be analyzed further to understand their behavior and contribution to language modeling tasks.

3.4.3 Gradient Activation

3.4.4 Contrastive Activation

Chapter 4

Data Collection and Extraction

4.1 Introduction

This chapter provides an overview of the initial survey, which encompasses survey site selection, data collection with the help of videography, pedestrian selection, and the extraction and coding of variables from the collected data.

4.1.1 Survey Site Selection

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4.1.2 Survey Data Collection

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4.1.3 Sample Selection

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4.1.4 Data Extraction and Coding

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4.2 Summary

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Chapter 5

Analysis and Results

5.1 Introduction

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5.2 Descriptive Statistics

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5.3 Model Development

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5.4 Discussion

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5.5 Summary

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Chapter 6

Conclusions and Practical Implications

6.1 Summary

This chapter outlines the findings and conclusions of the study, as well as the practical implications that can be drawn from the results.

6.2 Study Findings

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6.3 Research Contribution

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6.4 Practical Implications

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6.5 Study Limitations and Scope for Future Work

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Appendix A

Appendix A Title

Appendix B

Appendix B Title

References

List of Publications

Articles Published in International/National Journals

1. Article item 1
2. Article item 2

Presentations and Proceedings in International/National Conferences

1. Conference item 1
2. Conference item 2